



Review Article

Insights into the assessment and interpretation of earthquake-induced liquefaction in sands under different degrees of saturation

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ABSTRACT

Earthquake-induced liquefaction is a prominent and impactful natural hazard responsible for substantial economic losses worldwide. Hence, engineers and researchers are currently interested in developing methods and techniques to mitigate this destructive phenomenon. Reducing the degree of saturation is a reliable method to improve the liquefaction resistance of sandy soils since it directly influences the pore pressure build-up during seismic action. This paper reviews the mechanisms and assessment of earthquake-induced liquefaction in sandy soils with various degrees of saturation, a crucial parameter for reducing the phenomenon triggering. In addition, it presents novel approaches that delve into interpreting cyclic behaviour with diverse degrees of saturation using stress-based and energy-based approaches. The experimental results compiled and discussed show that, effectively, reducing the degree of saturation holds promise as a viable strategy for enhancing soil liquefaction resistance and mitigating associated risks. Moreover, the interpretation of cyclic behaviour addressed in this paper offers valuable insights into the reliability of interpreting methods to quantify the liquefaction resistance under several degrees of saturation (that may be achieved by desaturation or induced partial saturation techniques), contributing to strategies for resilience against earthquake-induced damages.

1. Introduction

Earthquakes are one of the most devastating natural phenomena (Kramer, 1996; Cremen and Galasso, 2020). Over the past century, numerous earthquakes have caused severe damage to buildings and infrastructure. Earthquakes have resulted in significant global economic losses, costing nearly EUR 27 billion (Boulanger and Bassal, 2024). Recent reports on liquefaction in Japan (Boulanger, 2012; Ishihara et al., 2016), New Zealand (Cubrinovski et al., 2011; Manzanal et al., 2021), Indonesia (Jaya et al., 2019; Watkinson and Hall, 2019), Philippines (Buhay et al., 2024) and Turkey (Bassal et al., 2024; Ersoz et al., 2024) have evidenced the risks to communities associated with liquefaction, providing an opportunity for improving public policies in various countries under similar risk.

One of the harmful phenomena associated with seismic action is earthquake-induced liquefaction. Although this phenomenon is not fatal, it has caused severe damage to buildings and infrastructure. The risks caused by earthquake-induced liquefaction are associated with a

significant reduction in the bearing capacity of the soil, large non-uniform ground deformation and lateral spreading (Jorge and Vieira, 1997; Grandin et al., 2007; Teves-Costa and Batlló, 2010; Bouziou and O'Rourke, 2017). These damages generally do not significantly affect the structure, but they cause the loss of serviceability of the constructions affected by the phenomenon (Viana da Fonseca et al., 2017; Lai et al., 2020). For example, from September 2010 to December 2011, Christchurch (New Zealand) experienced a series of earthquakes with Moment Magnitude (M_w) between 5.0 and 7.1, causing liquefaction and damaging more than 60,000 buildings (residential and commercial), essential facilities (i.e. electrical networks and pipelines) and hundreds of kilometres of roads in the region. In May 2012, Northern Italy was struck by a high-energy earthquake sequence (with M_w around 5.9) caused by soil liquefaction. Minarelli et al. (2022) describe liquefaction-induced damage after the 2012 Emilia earthquakes at 120 locations in the alluvial plain area between the Po, Reno and Secchia rivers, affecting the provinces of Bologna, Ferrara, Mantova and Modena. The liquefaction severely damaged critical infrastructure (e.g. hospitals, bridges and

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pipelines), old masonry buildings and newer constructions.

In this context, it is essential to identify mitigation techniques capable of alleviating damages caused by earthquake-induced liquefaction while adhering to standards for design reliability, environmental compatibility, and cost-effectiveness. Desaturation or Induced Partial Saturation (IPS) methods allow for the improvement of soil resistance against liquefaction by increasing the compressibility of the pore fluid by introducing a certain amount of air/gas. The presence of the air or gas creates occluded air bubbles in the pore fluid, reducing the degree of saturation (S_r), which varies depending on the technique employed. As the gas content of the fluid increases, the undrained strength and liquefaction resistance increase (Unno et al., 2008; Rebata-Landa and Santamarina, 2012; Kaminski and Grabe, 2023). It is widely documented that earthquake-induced liquefaction typically occurs in saturated soil deposits (Tatsuoka et al., 1986; Ishihara, 1993; He et al., 2014; Wang et al., 2021; Flora et al., 2022; Moug et al., 2022). Therefore, a reduction in the degree of saturation can increase the cyclic undrained shear strength (Wang et al., 2016; Flora et al., 2020; Banerjee et al., 2022; Fardad Amini and Yang, 2023).

This paper provides a comprehensive overview of the assessment of earthquake-induced liquefaction of sands with different degrees of saturation. It compiles a wide range of experimental results from the 1980s to 2024, particularly emphasising recent studies published in the last decade. These experimental results demonstrate how the degree of saturation affects the cyclic undrained shear strength and, thus, the effectiveness of desaturated or induced partial saturation techniques in mitigating earthquake-induced liquefaction. Therefore, this work significantly contributes to improving the understanding of the analysis and innovative methods for evaluating the cyclic behaviour of sandy soils under several degrees of saturation, particularly on two key aspects: (i) the influence of soil saturation on earthquake-induced liquefaction, and (ii) unconventional techniques for assessing liquefaction resistance under different saturation conditions. The methodologies discussed herein offer valuable insights into the effectiveness of desaturation soils to improve earthquake-induced liquefaction resistance and the reliability of interpreting methods for quantifying the cyclic behaviour improvement of sands under different S_r .

2. Influence of the degree of saturation on earthquake-induced liquefaction

The earthquake-induced liquefaction is an instability phenomenon characterised by the loss of the strength and stiffness of soil caused by seismic loading, which significantly reduces the effective stress due to the generation of pore pressure build-up (Ishihara, 1993). Once the soil liquefies, it presents characteristics and behaviour similar to that of a non-Newtonian fluid due to the viscosity variation in the fluid-saturated media (Lirer and Mele, 2019). Based on this context, liquefaction is one of the most complex phenomena in granular soils. Besides, it is one of the causes of severe damage to operational facilities and natural ecosystems due to the large displacements in the ground surface, including settlements and lateral spreading.

Any granular soil capable of generating pore pressure build-up is susceptible to trigger liquefaction; clean and uniform saturated sands are the soils most likely to liquefy (Boulanger and Idriss, 2014; Huang and Zhao, 2018; Viana da Fonseca et al., 2023). Since liquefaction is associated with the pore-pressure build-up, reducing the degree of saturation (S_r) increases the liquefaction resistance of soils. In natural deposits, the soil can present all or some of the following conditions: 'dry', 'unsaturated', 'partially saturated' and 'fully saturated' according to the volume of liquid (typically water) in the pores of the soil skeleton (Zeybek and Madabhushi, 2021). Unsaturated soils comprise three phases: solid, liquid and gas. The mechanical behaviour of unsaturated soils is characterised by capillary tension at the interface of the gas and liquid, generating negative pore pressure values due to matric suction (Fredlund, 2006). In the case of partially saturated soils, the voids are

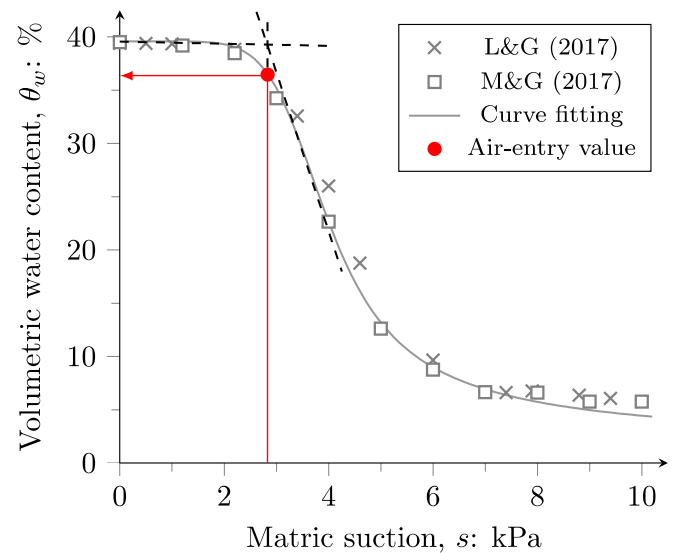


Fig. 1. Estimation of air entry value (Ottawa sand data from Le and Ghayoomi, 2017; Mirshekari and Ghayoomi, 2017).

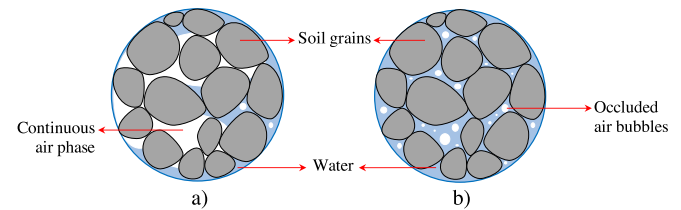


Fig. 2. Gas/air distribution in soils: a) unsaturated condition; b) gassy or partially saturated condition.

primarily filled with water, like in fully saturated soils. The difference between these conditions is that partially saturated soils have air/gas bubbles in the fluid (Molina-Gómez et al., 2023b). Despite being filled with water, occluded air bubbles mean these soils have no direct contact between the bubbles and the solid particles. As a result, partially saturated sands are sometimes referred to as 'gassy sands' due to this characteristic condition (Tarantino, 2013; Mele et al., 2024).

The threshold value of the S_r value that separates the partially or gassy state from the unsaturated conditions in sands is the air-entry value (AEV) (Fredlund, 2006; Molina-Gómez et al., 2023a). AEV can be determined by analysing the soil water characteristic curve, as it indicates the range of volumetric water content where the effects of matric suction become negligible. The volumetric water content (θ_w) is the ratio of the volume of water (V_w) to the total volume of the soil mass (V), as $\theta_w = \frac{V_w}{V} 100\%$. Matric suction (s) is the difference between the gas and liquid pressure within the soil pores. This pressure difference alters the stress state of the soil skeleton by increasing its effective stress (Fredlund, 2006).

Fig. 1 outlines the procedure for estimating AEV in granular soils using experimental data reported in the literature for Ottawa sand (Le and Ghayoomi, 2017; Mirshekari and Ghayoomi, 2017). This figure shows that AEV is represented by a specific s value (in kPa) for a given θ_w ; once the AEV is surpassed (or $s > \text{AEV}$), air or gas begins to interact with the soil particles, resulting in the development of capillary tension within the soil skeleton. For practical proposes, θ_w , which can also be expressed as the product of the soil porosity (n) and S_r , that is, $\theta_w = nS_r$. Hence, this allows for a representation of AEV to a particular S_r .

Zeybek and Madabhushi (2016) observed in partially saturated or gassy sands, the dimensions of air bubbles are typically smaller than the size of sand particles by using enlarged high-quality images. These

authors determined that these air bubbles exist discretely, fitting into the pore spaces without interacting with the soil grains. Consequently, the matric suction is negligible in partially saturated sandy soils, as pore water and air pressures are nearly equal (Zeybek and Madabhushi, 2021; Mele et al., 2024). This phenomenon occurs because occluded air bubbles are in equilibrium with the surrounding pore fluid. Fig. 2 illustrates the air distribution in unsaturated and partially saturated soils.

Two fundamental mechanisms cause the resistance improvement (Mousavi and Ghayoomi, 2021; Zeybek and Madabhushi, 2021; Kaminski and Grabe, 2023): [i] matric suction; and [ii] increase of fluid compressibility (or reduction of bulk modulus of pore fluid). Matric suction increases the effective stress, thus the soil stiffness and strength (Bishop and Blight, 1963; Unno et al., 2008). However, its contribution to liquefaction resistance is negligible since the maximum matric suction observed in clean and well-graded sands is lower than 5 kPa (Fredlund, 2006; Mirshekari and Ghayoomi, 2017; Cordeiro et al., 2022; Molina-Gómez et al., 2023a).

The occluded air bubbles increase the volumetric compressibility of the fluid-saturated media, reducing the pore pressure build-up during cyclic loading and, consequently, improving the liquefaction resistance (Okamura and Soga, 2006; Tsukamoto et al., 2014; Mele and Flora, 2019; Molina-Gómez et al., 2023b). Therefore, reducing the bulk modulus of fluid emerges as the primary mechanism that increases the liquefaction resistance of granular soils with S_r lower than 100 %. This is achieved by occluded air bubbles, which absorb the pore pressure build-up generated during seismic loading by reducing the soil volume (Pietruszczak et al., 2003; Okamura and Soga, 2006; Eseller-Bayat et al., 2013; Mele et al., 2019).

Therefore, soil desaturation or induced partial saturation (IPS) is considered one of the most innovative and promising methods for increasing liquefaction resistance (Pietruszczak et al., 2003; Marasini and Okamura, 2015; Zeybek and Madabhushi, 2018; Mousavi and Ghayoomi, 2021), once a certain amount of air or gas is introduced into the soil voids (Okamura et al., 2011; Eseller-Bayat et al., 2013; Le and Ghayoomi, 2017; Zeybek, 2022b). The introduction of air or gas reduces the degree of saturation and thus improves liquefaction resistance – for example, a reduction in saturation from 100 % to 95 % can lead to a 30 % increase in liquefaction resistance (Yegian et al., 2007; Jafarzadeh and Sadeghi, 2012; Mele and Flora, 2019; Molina-Gómez et al., 2023b).

Desaturation/IPS can be achieved in soil deposits using air (or gas) injection (Okamura et al., 2011; Marasini and Okamura, 2015; Zeybek and Madabhushi, 2016), water electrolysis (Yegian et al., 2007; Zhang et al., 2020), chemical gas generation (Eseller-Bayat et al., 2013;

Zeybek, 2022b), and biogenic gas generation processes (He et al., 2013; O'Donnell et al., 2019; Wang et al., 2021; Astuto et al., 2022; Moug et al., 2022; Zeybek, 2022b). The air in the voids prevents the soil from liquefying due in part to the effect of the matric suction (Zhang et al., 2016; Tsukamoto, 2019). Notwithstanding, the most crucial factor is the high-volumetric compressibility of air bubbles, even at the micro-scale (Okamura and Soga, 2006; Tsukamoto et al., 2014; Mele and Flora, 2019). The reduction in S_r leads to an increment of soil stiffness and a displacement of the critical state locus in non-plastic tailings due to matric suction effects (Robertson et al., 2017; Wang et al., 2023). The changes in locus position are not anticipated in clean and silty sands treated with desaturation techniques to induce gassy conditions, as the reduction in S_r primarily impacts pore pressure build-up rather than the critical state of these soils. However, this phenomenon presents a captivating research topic for future studies. During undrained cyclic loading, the matric suction generates an additional strength due to the increase in stress state, and the air/gas compression minimises the pore pressure build-up in granular soils due to the reduction in bulk modulus (Mele et al., 2020; Molina-Gómez et al., 2023b). The instability of granular soils under cyclic loading is more pronounced in the presence of non-compressible fluids (Lirer and Mele, 2019). This effect can be controlled by increasing the compressibility of the void phase with specific fluids or by injecting air/gas.

Recently, interest in studying geotechnical problems with desaturated soils has increased, especially in physical models for centrifuges (Caicedo and Thorel, 2014). Prototypes to assess the liquefaction effects in shallow foundations and attenuation techniques have shown settlement reduction after the desaturation of liquefiable layers (Dashti et al., 2010; Ghayoomi et al., 2013; Olarte et al., 2017). Air injection systems have successfully exhibited good results in desaturating the soil to mitigate liquefaction in the geotechnical centrifuge, as centrifuge testing showed that partial saturation influences the on-site response of sandy soil deposits (Marasini and Okamura, 2015; Mirshekari and Ghayoomi, 2017; Zeybek and Madabhushi, 2018). These results demonstrate the suitability of desaturation for liquefaction mitigation.

Once the procedure for estimating S_r and improving liquefaction resistance by varying S_r is established, the durability of desaturation should be investigated. However, very few studies have investigated the liquefaction susceptibility of natural soil deposits after applying the induced partial saturation technique. Yegian et al. (2007) observed a change in S_r from 82 % to 84 % over 442 days after desaturating the soil by water electrolysis. Okamura et al. (2011) collected high-quality undisturbed frozen samples from six experimental sites where the air injection technique was applied. Analysis of these samples in the laboratory revealed a slight increase in S_r content compared to values reported after the soil desaturation. These authors also conducted a series of numerical simulations. They concluded that the gas bubbles injected into the soil will be maintained for more than 26 years, demonstrating the long-term durability of soil resistance improvement. Nevertheless, the previously mentioned durability results are limited to hydrostatic conditions.

Moug et al. (2022) carried out a series of field trials to assess the performance of biogenic gas generation using a microbially induced desaturation method in liquefiable low-plasticity silty soil deposits. The performance of these field trials was evaluated by V_p in situ measurements. These results successfully demonstrated the biogenic gas generation performance in desaturating liquefiable soils, reporting S_r lower than 98 %, in which the desaturation persisted for at least 92 days after treatment. Hu et al. (2020) conducted a series of tests under hydraulic gradient flow to assess the stability of biogas generated by bacterial denitrification. In the seepage condition, S_r changed from 84.7 % to 85.5 %. No biogas bubble aggregation occurred, and no biogas bubbles were observed to flow away from the sample. Hence, these authors attributed the decrease in biogas bubbles to the gas dissolution caused by seepage. The performance of desaturation techniques in hydrostatic and seepage conditions is not significantly affected by the void ratio (or

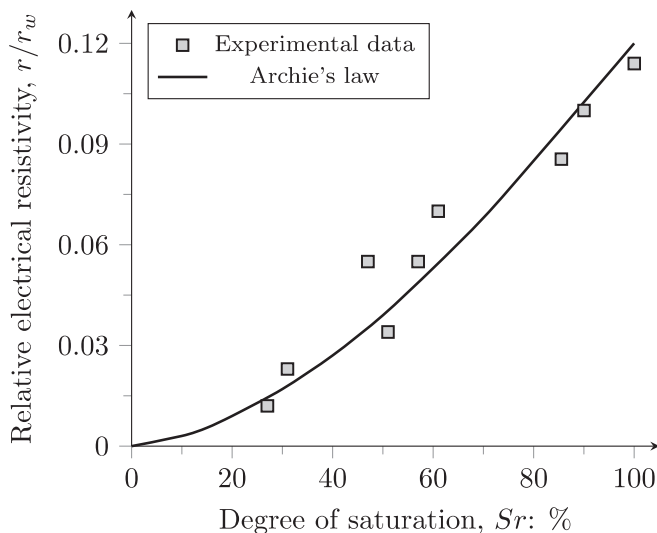


Fig. 3. Relation between relative electrical resistivity and degree of saturation based on Archie's theory (after Cosentini and Foti, 2014).

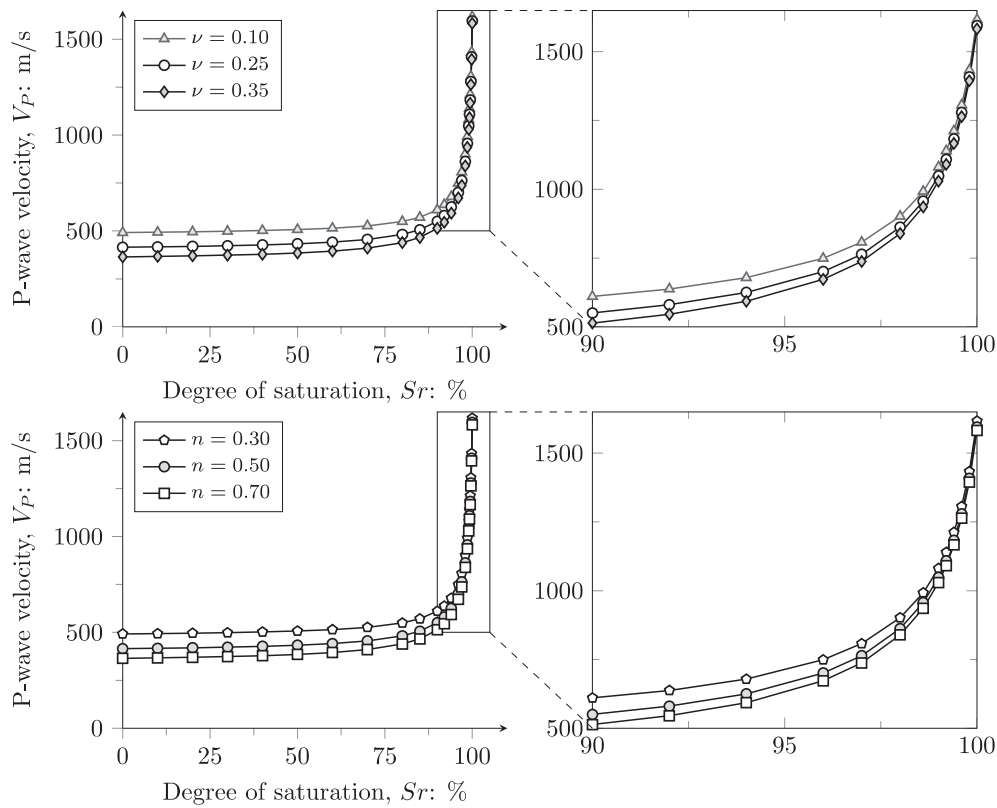


Fig. 4. Relation between P-wave velocity and degree of saturation based on Biot's theory (after Santamarina et al., 2005): a) evolution for different Poisson's ratios; b) evolution for different porosities.

relative density) of the soil (Zeybek and Madabhushi, 2017; Mousavi et al., 2021).

Although few studies evaluate the durability of the desaturation technique and whether air bubbles remain in the field, it is necessary to find a way to reliably measure S_r during the operation of any IPS technique, which is critical for evaluating its performance. In the laboratory, the B -value is the most common parameter for estimating S_r . However, this parameter does not provide a reliable estimate of S_r in the field, as in free field conditions, the undrained condition cannot be guaranteed. Therefore, alternative methods involving electrical conductivity or P-wave velocity (V_p) have demonstrated promising results in estimating S_r by both in situ and laboratory testing (Valle-Molina and Stokoe, 2012; Cosentini and Foti, 2014; Chen et al., 2021; Astuto et al., 2022; Moug et al., 2022; Molina-Gómez et al., 2024). Electrical resistivity (r) and P-wave velocity in soils depend on the saturation and other parameters, such as the porosity, fabric, stress, mineralogy, pore fluid and temperature (Santamarina et al., 2005). Hence, V_p and relative electrical resistivity (i.e. the ratio between electrical resistivity and electrical resistivity of water, r/r_w) can be used as reliable estimators of S_r .

Fig. 3 and Fig. 4 show the theoretical approaches based on the electrical resistivity and wave-propagation theories (Archie, 1942; Biot, 1956), respectively. Fig. 3 exhibits results of relative electrical resistivity, r/r_w , for various S_r reported by Cosentini and Foti (2014). Fig. 4 presents an example of the evolution V_p as a function of S_r for sands with different values of porosity (n) and Poisson's ratio (ν) based on data reported by Santamarina et al. (2005). As observed in Fig. 3 and Fig. 4, V_p and r/r_w demonstrate significant potential for evaluating S_r in soil deposits. These parameters offer a robust theoretical framework enabling monitoring saturation levels in laboratory and field conditions, thus providing valuable insights into the effectiveness and long-term durability of desaturation or IPS techniques.

Additionally, wave propagation and electrical resistivity

measurements can be employed to assess the cyclic behaviour of sands with varying degrees of saturation. These laboratory techniques facilitate the evaluation of liquefaction resistance and help understand the interaction between soil properties and saturation levels. Hence, integrating these methods allows for predicting how varying S_r impacts liquefaction resistance, ensuring a comprehensive evaluation of soil stability under seismic loading.

3. Interpretation of liquefaction resistance under cyclic loading for diverse degrees of saturation

Eurocode 8 (EN 1998-5, 2010) recommends the simplified procedure proposed by Seed et al. (1975) to define the earthquake-induced loading, considering the seismic wave propagation in a soil column. In the last decades, further procedures have been proposed to estimate the liquefaction susceptibility from diverse approaches based on different soil parameters (Robertson and Wride, 1998; Andrus and Stokoe II, 2000; Juang et al., 2002; Cetin et al., 2004; Moss et al., 2006; Idriss and Boulanger, 2008; Kayen et al., 2013; Boulanger and Idriss, 2014). These cases provided data to propose correlations between liquefaction triggering and in situ test results involving SPT, CPT and seismic soundings (or S-wave velocity) (NASEM, 2021). The methods also include parameters such as the vertical effective stress σ'_v , the fines content, and the equivalent cyclic stress ratio for an equivalent earthquake of $M_w = 7.5$ ($CSR_{M7.5}$) estimated by different in situ test methods.

As indicated above, the assessment uses in situ test results for a specific seismic action. The in situ tests for liquefaction assessment can be reliably complemented in the laboratory by testing undisturbed samples with high quality collected by advanced sampling techniques (Molina-Gómez et al., 2020). High-quality sample testing is an excellent alternative, but these samples are only used for research studies or relevant infrastructure projects (e.g. bridges or nuclear power plants). Establishing the earthquake-induced liquefaction assessment of natural

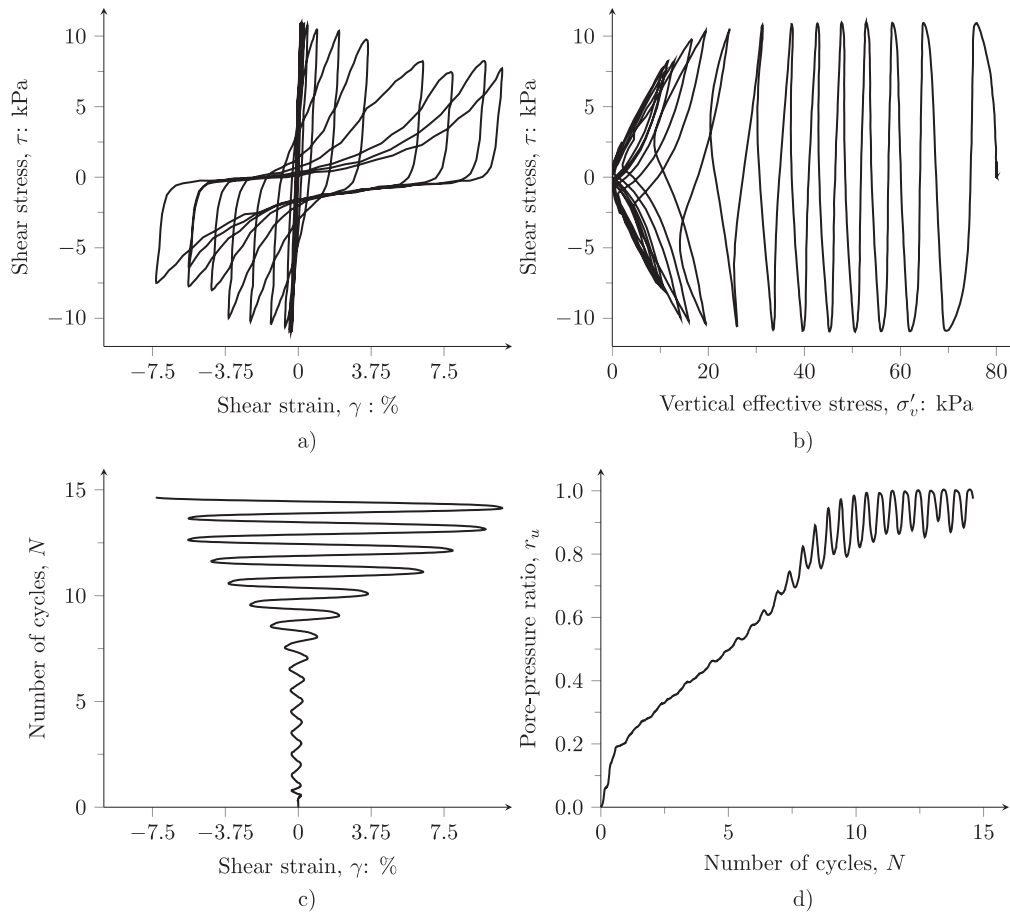


Fig. 5. Cyclic soil response – experimental data of loose sand during direct simple shear testing (Viana da Fonseca et al., 2023): a) Stress-strain cyclic behaviour; b) Stress path; c) Strain evolution; d) Pore pressure build-up.

soil deposits on laboratory results is non-conventional in current design practice for diverse reasons, emphasising the difficulty of undisturbed sample testing and the limitation of applying uniform cyclic loading patterns. However, laboratory studies on reconstituted soils are beneficial for studying behaviour patterns or validating analysis methods, such as the effects of soil desaturation in liquefaction resistance assessment. Therefore, the results of this study cover experimental data obtained from remoulded samples subjected to uniform cyclic loading conditions. The following sections address assessing liquefaction resistance under different degrees of saturation by laboratory testing.

3.1. Conventional assessments of liquefaction resistance

In the laboratory, the number of cycles for liquefaction onset (N_L) is used to characterise the liquefaction resistance of soils under a representative earthquake motion by uniform loading in laboratory testing. This approach involves the assessment of N_L for different combinations of cyclic stress ratio (CSR) by element testing (see Fig. 5). During element testing, a soil sample is subjected to an undrained cyclic loading to measure the evolution of shear strain and pore pressure build-up. The CSR comprises a series of uniform loading cycles representing a series of irregular cyclic loads representative of a specific seismic action (Seed et al., 1975; Ishihara, 1993; Kramer, 1996).

CSR is the applied cyclic stress divided by the initial effective confining stress, representing an earthquake time history. Therefore, CSR computation depends on the nature of loading during element testing, e.g. axial, torsional, or tangential for triaxial, torsional, hollow cylinder or simple shear apparatus, respectively. The assessment of liquefaction resistance based on the pore pressure build-up is known as

the stress criterion and uses the pore-pressure ratio (r_u). The liquefaction occurs when the pore pressure build-up is near or equal to the initial effective stress of the soil, leading to r_u values from 0.90 to 1.0 to define the liquefaction onset, as indicated by different authors (Polito et al., 2008; Jefferies and Been, 2016; Mele et al., 2019; Viana da Fonseca et al., 2023). Simultaneously, the soil deforms progressively, meanwhile the pore pressure build-up approaches the effective confining pressure during cyclic loading (Ishihara, 1996). Such an evolution of strains provides another liquefaction criterion, the strain criterion. The strain criterion considers the liquefaction onset from a specific double strain amplitude (DA) value, which changes according to the loading type. Table 1 shows the procedures to compute CSR and r_u , together with the DA value for different testing methods and apparatuses in the laboratory.

The tests listed in Table 1 replicate specific testing conditions. In cyclic triaxial tests, compression and extension deviatoric stresses induce an inversion of principal stresses. For cyclic torsional shear, cyclic simple shear, and cyclic direct simple shear, the distortional loading results in the rotation of principal stresses. The hollow cylinder apparatus allows the application of multiaxial cyclic loading that both inverts and rotates principal stresses simultaneously. In all tests in Table 1, direct measurements of pore pressure build-up can be conducted, except in the direct simple shear test. In such a test, the effective stress variation can be estimated by measuring the reduction in vertical loading or stress under a constant volume condition.

In element testing, the incremental pore pressure build-up and evolution of strains are measured along with the number of loading/stress cycles. The test results for liquefaction assessment cover the number of cycles for liquefaction onset (N_L) for each CSR. Such a number of cycles

Table 1
CSR for different cyclic element testing.

Type test	CSR	r_u	DA value*
Cyclic triaxial	$CSR = \frac{q}{2p_0}$	$r_u = \frac{\Delta u}{p_0}$	$e_a = 5.0\%$
Cyclic torsional shear	$CSR = \frac{\tau}{p_0}$	$r_u = \frac{\Delta u}{p_0}$	$\gamma = 7.5\%$
Hollow cylinder	$CSR = \frac{\tau}{p_0}$	$r_u = \frac{\Delta u}{p_0}$	$\gamma = 7.5\%$
Cyclic simple shear	$CSR = \frac{\tau}{p_0}$	$r_u = \frac{\Delta u}{p_0}$	$\gamma = 7.5\%$
Cycle direct simple shear	$CSR = \frac{\tau}{\sigma'_0}$	$r_u = \frac{\Delta u}{\sigma'_0}$	$\gamma = 7.5\%$

* Criteria from Ishihara (1996).

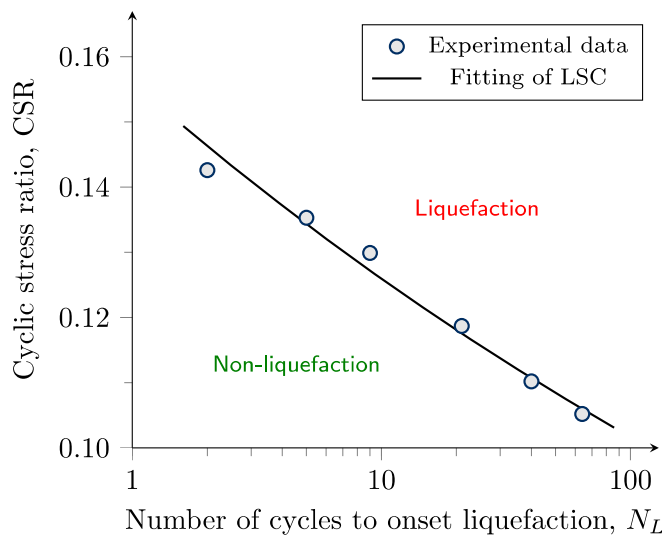


Fig. 6. Schematic of liquefaction strength curve –data from Viana da Fonseca et al. (2023).

can be estimated by considering the r_u or DA strain criteria.

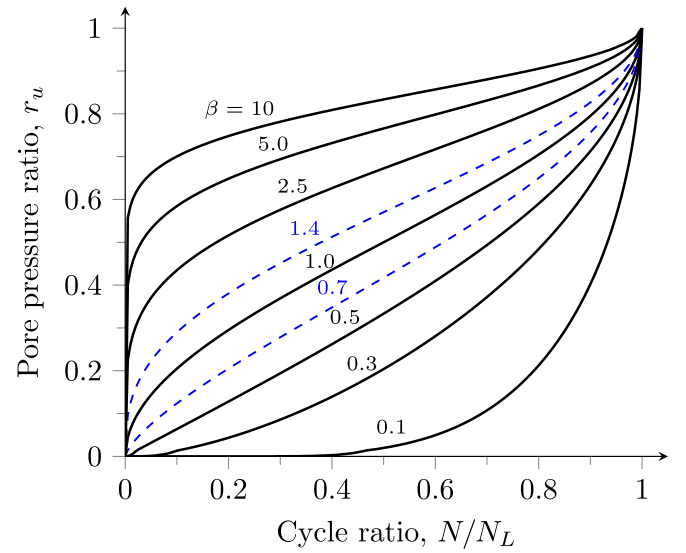


Fig. 7. Evolution of r_u evolution as a function of N/N_L – dashed lines denote the boundaries recommended by Booker et al. (1976).

$$\beta = C_1 \bullet FC + C_2 \bullet Dr + C_3 \bullet CSR + C_4 \quad (7)$$

Experimental data provide a series of points that compose the liquefaction strength curve (LSC). Fig. 6 illustrates the LSC from experimental data, represented in a semi-logarithmic space of CSR vs. N_L , showing that CSR above LSC designates liquefaction manifestation. In contrast, CSR below LSC designates non-liquefaction onset.

The LSC curve depends on the soil state (e.g. confinement, relative density, or degree of saturation). Hence, an LSC only represents the cyclic behaviour of a specific degree of saturation (S_r). Usually, the LSC follows a power rule in the form of Eq. 5 (Boulanger and Idriss, 2014).

$$CSR = a(N_L)^{-b} \quad (5)$$

where a and b are fitting parameters, representing the intercept and slope, respectively, of the linear projection in the CSR – $\log N_L$ space. These parameters are derived from regression analysis of the experimental data. Moreover, they can change with diverse physical factors affecting the liquefaction susceptibility of sands, highlighting the state parameter and the degree of saturation of the soil.

3.2. Stress-based approach to estimate liquefaction resistance

In the case of clean sands, initiating liquefaction can unequivocally be considered a state of softening induced by significant deformations, leading to sudden and complete loss of strength occurring during or immediately following the point where $r_u \approx 1$. Therefore, the double strain amplitude (DA) or pore pressure build-up criteria can identify the liquefaction onset since the earthquake-induced liquefaction is associated with large strains caused by a reduction in effective stress (Ishihara, 1993). However, the soil response under cyclic loading depends on the state, e.g. effective stress, relative density, and degree of saturation (Boulanger and Idriss, 2014; Viana da Fonseca et al., 2015; Molina-Gómez et al., 2019). Consequently, the selection of criteria to define the onset of liquefaction is a decision that depends on the soil type and testing conditions.

Due to the relevance of the stress criterion, several authors have explored models to predict the pore pressure build-up in granular soils (Lee and Albaisa, 1974; Seed et al., 1975; Booker et al., 1976; Wang and Kavazanjian, 1989; Polito et al., 2008; Park et al., 2014; Chiaradonna and Flora, 2020). These models were proposed by fitting experimental

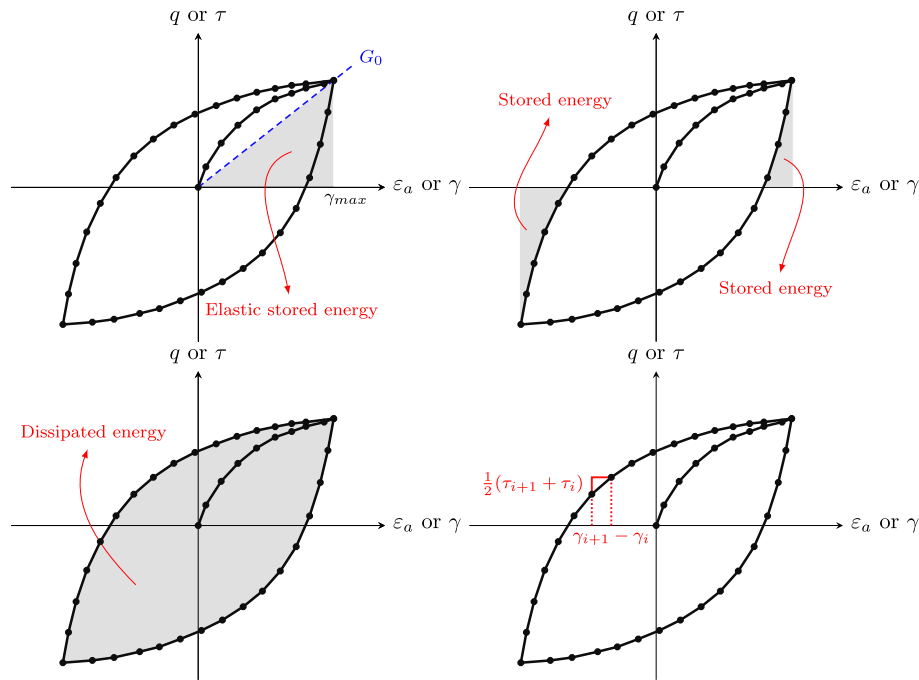


Fig. 8. Schematic representation of energy computation from experimental data.

data of cyclic element tests and following the formulation developed by Seed et al. (1975) to predict the r_u as a function of the cycle ratio, N/N_L . Such a formulation uses a coefficient β , which depends on the type and state of the soil. Eq. 6 describes the formulation proposed by Seed et al. (1975).

$$r_u = \frac{1}{2} + \frac{1}{\pi} \sin^{-1} \left[2 \left(\frac{N}{N_L} \right)^{1/\beta} - 1 \right] \quad (6)$$

From Eq. 6, it can be inferred that β controls the shape of the curve describing r_u evolution. Booker et al. (1976) recommended $\beta = 0.7$ and $\beta = 1.4$ as lower and upper boundaries, respectively, for clean sands. Fig. 7 shows the r_u evolution as a function of N/N_L and compares different β values. Note that dashed lines denote the boundaries recommended by Booker et al. (1976). Wang and Kavazanjian (1989) showed that β is a function of the cyclic stress ratio (CSR) and developed a procedure for predicting pore pressure development in non-uniform loading using the curves relating pore pressure ratio to N/N_L as a function of CSR. Due to the association of β with soil type and testing conditions, Polito et al. (2008) introduced a correlation relating this coefficient to fines content

(FC), relative density (D_r) and cyclic stress ratio (CSR), as follows in Eq. 7, where C_1 , C_2 , C_3 and C_4 are regression coefficients. Polito et al. (2008) indicated that for soils with $FC < 35\%$ $C_1 = 0.01166$, $C_2 = 0.007397$, $C_3 = 0.01034$ and $C_4 = 0.5058$; while for soils with $FC \geq 35\%$ $C_1 = 0.002149$, $C_2 = -0.0009398$, $C_3 = 1.667$ and $C_4 = 0.4285$.

Integrating models that predict pore pressure build-up into a coupled dynamic numerical analysis is challenging (Park et al., 2014; Gao et al., 2023; Moreno-Torres et al., 2023). The above is because N/N_L must be specified a priori, and such analysis seeks to establish the stability, deformations and possible damages induced by pore pressure increments in geotechnical structures (Chiaradonna et al., 2018). However, models such as the PM4Sand model, the hypoplastic model with intergranular strain, the intergranular strain anisotropy model, and the UBSCAND model have been implemented to reproduce the effects and the accumulation of pore water pressure on the mechanical behaviour of the sands under undrained cyclic loading at different stress/strain amplitudes (e.g. Ziotopoulou and Boulanger, 2016; Wichtmann et al., 2019; ElGhoraiby and Manzari, 2021; Duque et al., 2023).

3.3. Energy-based approach to estimate liquefaction resistance

Energy-based methods for evaluating soil liquefaction were introduced as an alternative to stress-based procedures by Nemat-Nasser and Shokooh (1979) for analysing historical earthquake cases. This energy approach was proposed based on seismological application to estimate the soil densification for drained conditions and pore pressure build-up for undrained conditions. Besides, it is based on the idea that the soil deforms as it dissipates part of the energy during cyclic loading (Figueroa et al., 1994). The energy-based approaches have been applied to predict the liquefaction susceptibility of natural soil deposits with great success by Kokusho (2013) and validated numerically by Millen et al. (2020).

The energy-based procedures involve various measures of energy, including the elastic stored energy (W), stored energy (W_s) and total dissipated energy per unit volume (ΔW), which is appropriate since cyclic response involves energy transformation (Liang, 1995; Green et al., 2000). The dissipated energy per unit volume of all cycles before liquefaction onset is linked to the pore pressure build-up for liquefaction assessment (Khashila et al., 2021). Fig. 8 schematises the graphical

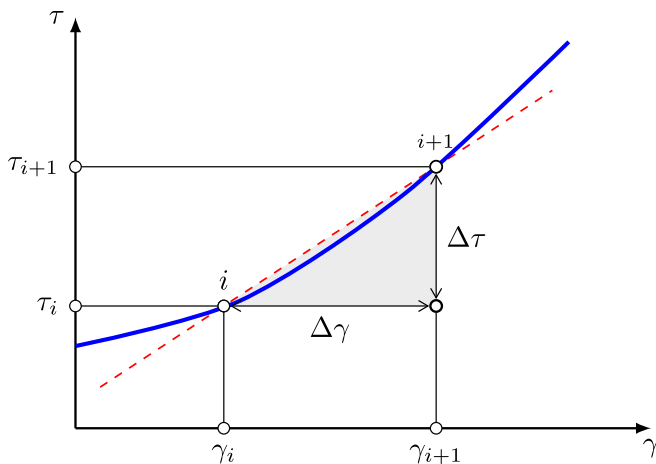


Fig. 9. Numerical integration to compute the energy during cyclic loading.

definition of W , W_r and ΔW for one stress-strain hysteresis loop.

Eq. 8 allows quantifying (ΔW) by relating the stresses and incremental strains (note that both the stress and incremental strain tensors are symmetrical), as follows:

$$\Delta W = \int dW = \int \sigma'_x d\epsilon_x + \sigma'_y d\epsilon_y + \sigma'_z d\epsilon_z + \tau_{xy} d\gamma_{xy} + \tau_{xz} d\gamma_{xz} + \tau_{yz} d\gamma_{yz} \quad (8)$$

where dW is the incremental dissipated energy per unit volume; σ'_x , σ'_y , and σ'_z are the effective stresses acting parallel to the x , y and z axes, respectively; $d\epsilon_x$, $d\epsilon_y$ and $d\epsilon_z$ are the incremental axial strains (ϵ) in the x , y and z directions, respectively; τ_{xy} , τ_{xz} and τ_{yz} are the shear stresses acting on planes having normal vectors in the x , y and z directions, respectively; incremental shear strains $d\gamma_{xy}$, $d\gamma_{xz}$ and $d\gamma_{yz}$ resulting from the shear strains (γ) on the normal planes of vectors in the x , y and z directions.

Eq. 8 can be solved by numerical integration (e.g. using the trapezoidal rule), as illustrated in Fig. 9. The numerical solution allows deriving the normalised unit energy (W_s), the parameter used for liquefaction assessment (Green et al., 2000; Polito et al., 2013; Khashila et al., 2021). Eqs. 9, 10 and 11 present the numerical derivation of Eq. 8 for estimating W_s from experimental data obtained from cyclic triaxial, cyclic torsional (for solid and hollow soil specimens) and cyclic direct simple shear tests, respectively.

$$W_s = \frac{1}{2p'_0} \sum_{i=1}^{n-1} (q_i + q_{i+1})(\epsilon_{a,i} - \epsilon_{a,i+1}) \quad (9)$$

$$W_s = \frac{1}{p'_0} \sum_{i=1}^{n-1} (\tau_i + \tau_{i+1})(\gamma_i - \gamma_{i+1}) \quad (10)$$

$$W_s = \frac{1}{\sigma'_0} \sum_{i=1}^{n-1} (\tau_i + \tau_{i+1})(\gamma_i - \gamma_{i+1}) \quad (11)$$

where i represents the load/stress increment, n is the total number of increments to achieve the liquefaction onset; q and τ are deviatoric and shear stresses, respectively; ϵ_a and γ are axial and shear strains, respectively; and p'_0 are σ'_0 initial mean effective stress and initial vertical stress, respectively.

Various authors, such as Figueroa et al. (1994), Azeiteiro et al. (2017) and Khashila et al. (2018), have observed that the magnitude of strain and the position of the peak loading cycle do not impact the outcomes of energy-based methods. Since the dissipated energy required for liquefaction onset is independent of the applied loading pattern (uniform or non-uniform), energy-based methods are up-and-coming for liquefaction assessment (Millen et al., 2020; Mele et al., 2023). These methods allow for comparing the results from different laboratory testing procedures and predicting strains and pore pressure generation in the field (Kokusho, 2013; Galindo-Aires et al., 2019). Despite their potential, these methods have yet to be widely adopted in practice and remain the focus of a relatively small fraction of the geotechnical earthquake engineering research community.

Another advantage of energy-based methods is the association between the energy applied during cyclic loading and pore pressure build-up. Green et al. (2000) proposed a model, which provides an empirical expression for relating r_u to the W_s of soil and a calibration parameter denominated as the pseudo-energy capacity (PEC), see Eq. 12. Such a model was developed using experimental data from cyclic element tests on non-plastic silt-sand mixtures, which ranged in fines from clean sands to pure silts.

$$r_u = \sqrt{\frac{W_s}{\text{PEC}}} \quad (12)$$

The calibration parameter PEC is estimated from cyclic test data by

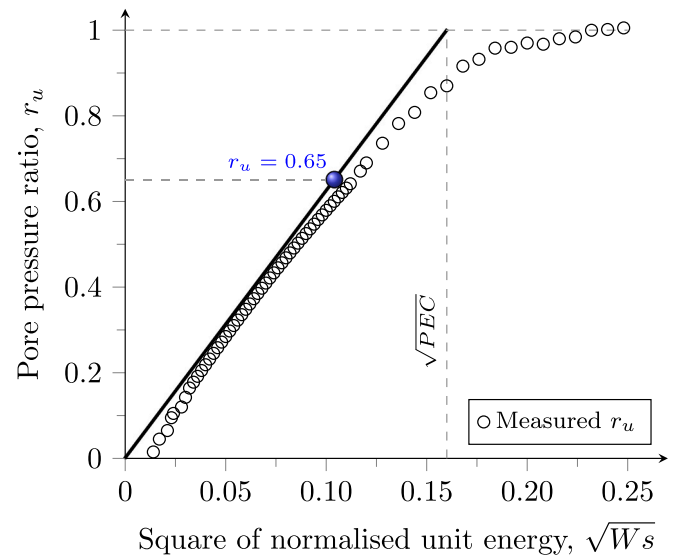


Fig. 10. Graphical illustration of PEC estimation follows (adapted from Green et al., 2000). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Reported logarithmic energy-based models for liquefaction assessment (after Alavi and Gandomi, 2012).

Reference	Model	Testing apparatus	Testing method	Soil
Figueroa et al. (1994)	$\log W_s = 2.002 + 0.0048p'_0 + 0.0116Dr$	Cyclic torsional shear	Stress control	Reid Bedford sand
Liang (1995)	$\log W_s = 2.062 + 0.0039p'_0 + 0.0124Dr$	Cyclic torsional shear	Strain control	Reid Bedford sand
	$\log W_s = 2.484 + 0.0047p'_0 + 0.0005Dr$	Cyclic torsional shear	Strain control	LSFD sand
Figueroa et al. (1999)	$\log W_s = 2.055 + 0.0048p'_0 + 0.0127Dr$	Cyclic torsional shear	Strain control	LSI-30 sand
	$\log W_s = 1.330 + 0.0061p'_0 + 0.0219Dr$	Cyclic torsional shear	Stress control	Nevada sand
Dief and Figueroa (2001)	$\log W_s = 2.459 + 0.0045p'_0 + 0.0012Dr$	Geotechnical centrifuge	60 g	Reid Bedford sand
	$\log W_s = 1.1164 + 0.0124p'_0 + 0.0124Dr$	Geotechnical centrifuge	60 g	Nevada sand

Note: W_s is the accumulated energy required for liquefaction onset (in J/m^3); p'_0 is the initial mean effective stress; Dr is the relative density.

plotting r_u versus the square root of W_s or calculated as follows (Green et al., 2000):

$$\text{PEC} = \frac{W_{S_{r_u=0.65}}}{(0.65)^2} = \frac{W_{S_{r_u=0.65}}}{0.4225} \quad (13)$$

The term pseudo-energy capacity or PEC commonly indicates the physical significance of the calibration parameter, being more than a fit parameter (Polito et al., 2008). Consequently, PEC represents the normalised unit energy dissipated at the point of the liquefaction onset (Green and Mitchell, 2012). Fig. 10 presents a graphical illustration of the PEC estimation.

Energy-based methods offer a robust approach for evaluating

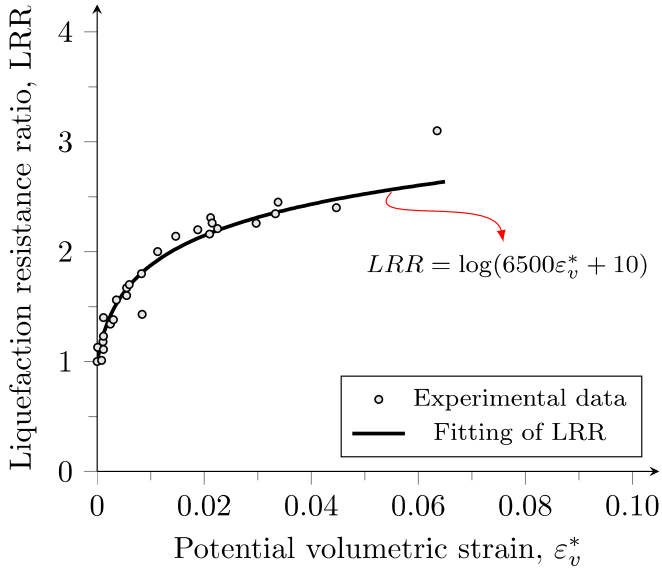


Fig. 11. Potential volumetric strain vs liquefaction resistance ratio (adapted from Okamura and Soga, 2006).

earthquake-induced liquefaction across various contexts, including historical cases, in situ testing, cyclic element testing, and physical modelling. These methods address parameters to quantify both the load imposed on the soil by the earthquake and the capacity needed to induce liquefaction, contrary to stress-based and strain-based methods (Alavi and Gandomi, 2012). Therefore, several models involving an energy-based approach have been developed to relate the soil energy for the liquefaction onset of sands with the shear strain amplitude and the state of soil (e.g. mean effective stress and relative density).

Table 2 summarises a series of logarithmic regression models to estimate the accumulated energy required for liquefaction onset as a function of the mean effective stress and relative density for diverse sands and based on different testing procedures, including element testing and physical models. This table shows that the logarithmic regression models do not consider the influence of S_r on liquefaction resistance, and they are not correlated with these energy functions. This observation underscores the need to further explore the S_r effects in centrifuge testing, as limited studies are available in the literature (Ghayoomi et al., 2013; Marasini and Okamura, 2015; Mirshekari and Ghayoomi, 2017; Hall et al., 2018; Zeybek and Madabhushi, 2018; Kumar et al., 2019).

Okamura and Soga (2006) proposed a method for assessing the effect of gas compressibility on liquefaction resistance by estimating the relationship between the liquefaction resistance ratio (LRR) with the potential volumetric strain (ϵ_v^*) achieved at $r_u = 1$. These authors defined LRR as the normalising of liquefaction resistance ratio obtained in gassy or partially saturated conditions in terms of the liquefaction resistance in fully saturated conditions, i.e. $LRR = CSR_g/CSR_s$. Potential volumetric strain, ϵ_v^* , represents the capacity of a soil to absorb the pore pressure build-up by reducing its volume and is estimated through Eq. 14:

$$\epsilon_v^* = \frac{p'_0}{BP + p'_0} (1 - S_r) \frac{e}{1 + e} \quad (14)$$

where p'_0 corresponds to the initial mean effective stress, BP denotes the back pressure during shearing (also designated as absolute fluid pressure), S_r is the degree of saturation, and e is the void ratio of the soil. Such a method involved the compilation of triaxial test results conducted in Toyoura sand under different effective confinement stresses (from 50 kPa to 200 kPa), relative densities (from 40 % to 90 %) and degrees of saturation (from 70 % to 100 %). Fig. 11 presents the relation between potential volumetric strains with the liquefaction resistance

ratio proposed by Okamura and Soga (2006).

For sands with S_r lower than 100 %, the total specific energy of deformation (W_{tot}) approach is used to assess the liquefaction susceptibility (Mele and Flora, 2019). The main feature of the W_{tot} approach is the incorporation of volumetric deformations during cyclic loading, which are mainly controlled by the fluid compressibility –that is, gas and liquid mixture. Okamura and Soga (2006) proposed a theoretical formulation to relate the volumetric strain at liquefaction onset ($\epsilon_{v,liq}$) with the initial degree of saturation (S_{r0}) based on Boyle's law. Such a formulation also involves the initial void ratio (e_0), the initial air pressure ($u_{a,0}$) and the initial effective confinement stress (p'_0). Eq. 15 describes the model to compute the $\epsilon_{v,liq}$ of partially saturated soils.

$$\epsilon_{v,liq} = \frac{e_0}{1 + e_0} (1 - S_r) \left(1 - \frac{u_{a,0}}{p'_0} \right) \quad (15)$$

For $S_{r0} = 1.0$, the volumetric strain is zero. In such a case, the energy-based methods in Fig. 8 reasonably estimate liquefaction susceptibility. However, to apply Boyle's law to partially saturated soils, the system must follow the subsequent thermodynamic hypotheses (Mele et al., 2019):

- The process is isothermal, meaning no heat is generated or lost during the test,
- The mass of the system remains constant, with no increase or decrease in the mass of air, water, or soil in the specimen during the test,
- The system is thermodynamically open, implying that the deformation process involves internal flows of air and water within the specimen,
- The pore gas (air) can be treated as an ideal gas.

Under the hypotheses of Boyle's law, Li (2007) proposed the following model based on Helmholtz function (Ψ) to represent the gradient $d\Psi$ as a function of the internal energy during cyclic loading under constant temperature, adopting the form:

$$d\Psi = [(p' - u_a\delta) + sSr\delta] : d\epsilon - \frac{e}{1 + e_0} s dS_r + \frac{e_0}{1 + e_0} (1 - S_{r0}) u_a d(\ln p_a) \quad (16)$$

where the term $(p' - u_a\delta)$ is the net stress, u_a is the pore air pressure, s is the suction, S_r is the degree of saturation, e is the void ratio of the soil, the subscript '0' indicates the initial value, and ρ_a is the mass density of the air. According to Li (2007), the incremental dissipation is neglected.

Mele et al. (2019) incorporated the Helmholtz function (Eq. 16) in the concept of total specific energy to assess the liquefaction resistance in soils with S_r lower than 100 %. Such a new energy-based approach considers the sum of three components: [i] the work associated with the deformation of the soil skeleton, [ii] the work caused by the flow of water mass, and [iii] the work induced by the air into the porous media. Hence, the total work (W_{tot}) can be expressed as:

$$W_{tot} = W_{sk} + W_w + W_a \quad (17)$$

where W_{sk} , W_w and W_a are the specific work that causes deformations of the soil skeleton, water, and air, respectively. W_{sk} covers the work due to volume change ($W_{v,sk}$) and the work caused by deviatoric strains ($W_{d,sk}$), as follows in Eq. 18:

$$W_{sk} = W_{v,sk} + W_{d,sk} \quad (18)$$

Considering the measured volumetric strains during element testing and incorporating thermodynamic hypotheses of Boyle's law, the elements of Eqs. 17 and 18 can be evaluated as:

$$W_{v,sk} = \int_0^{\epsilon_{v,liq}} [(p' - u_a) + sSr] d\epsilon_v \quad (19)$$

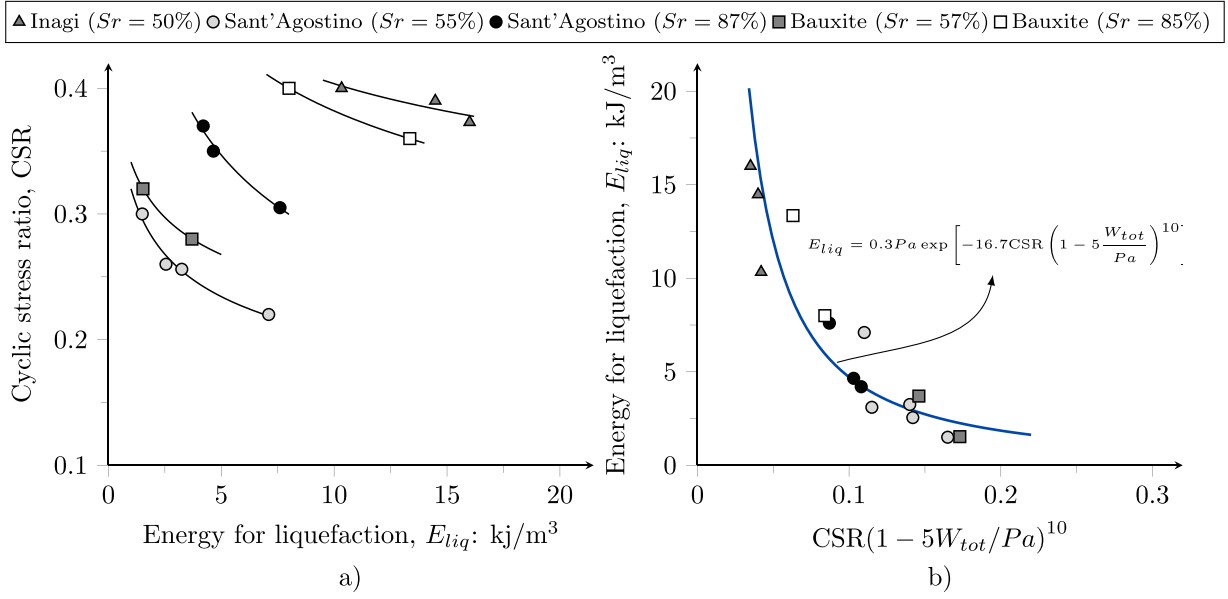


Fig. 12. Application of energy-based method for liquefaction resistance assessment in soils with different degrees of saturation (adapted from Mele and Flora, 2019): a) Energy for energy liquefaction under different CSR; b) Relation between the energy for energy liquefaction and the term $CSR(1 - 5W_{tot}/Pa)^{10}$.

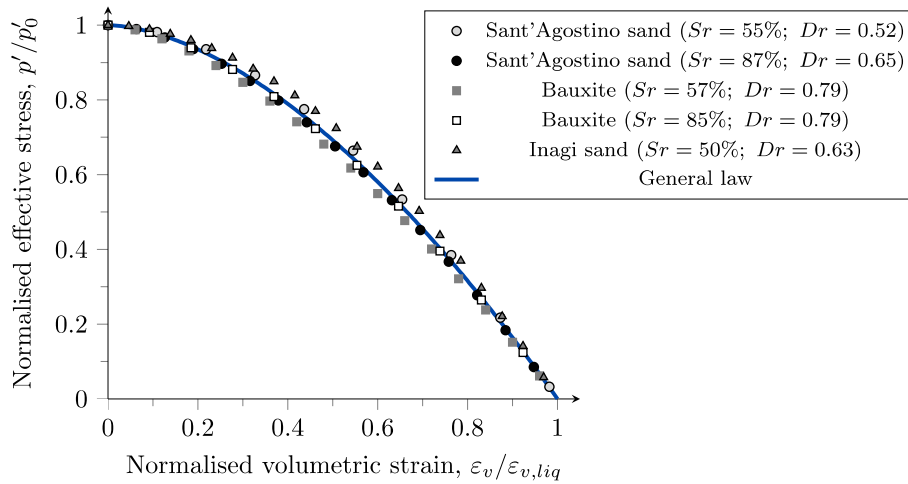


Fig. 13. General law of normalised effective stress against the normalised volumetric strain of soils with $Sr \leq 100\%$ subjected to cyclic loading (adapted from Mele and Flora, 2019).

$$W_{d,sk} = \iint_{\Delta W} dq d\varepsilon_d \quad (20)$$

$$W_w = - \int_{Sr_0}^{Sr_{liq}} \frac{e(Sr)}{1 + e(Sr)} s(Sr) dSr \quad (21)$$

$$W_a = \frac{e_0}{1 + e_0} (1 - Sr_0) u_{a,liq} d(\ln p_{a,liq}) \quad (22)$$

Mele et al. (2019) stated that $W_{v,sk}$ depends on the soil state (i.e. p'_0 , e_0 and Sr) and is independent of CSR, while $W_{d,sk}$ only comprises the dissipated strain energy. The energy for liquefaction onset (E_{liq}) depends on the work caused by deviatoric strains ($W_{d,sk}$) in each cycle (Mele and Flora, 2019). Hence, E_{liq} physically represents the sum of $W_{d,sk}$ across all cycles necessary for the phenomenon triggering, which can be estimated from the sum of the areas of all loading cycles (as indicated in Fig. 8). Besides, W_w is associated with the liquefaction of soils with diverse Sr , attributing volumetric deformations during cyclic loading to the compressibility of air/gas in the pore fluid. Lastly, W_a describes the

effect of pressure variation in the gas phase.

Fig. 12 exemplifies the energy-based method applied to predict the liquefaction resistance in soils with different Sr . Fig. 12a presents experimental results for deriving the evolution of E_{liq} under various CSR. This figure demonstrates that the relation between E_{liq} and CSR leads to a series of curves like those observed in the correlation of N_L with CSR, which also depend on the tested soil and Sr . However, Mele and Flora (2019) derived a nonlinear relationship linking E_{liq} to the term $CSR(1 - 5W_{tot}/Pa)^{10}$ (see Fig. 12b). This relationship provides a more general interpretation, describing the behaviour of soils with different Sr , independent of soil type and testing conditions. These results validate the effectiveness of energy-based approaches in assessing liquefaction resistance under diverse Sr .

Mele et al. (2019) and Mele and Flora (2019) observed that the Boyle law (see Eq. 15) reveals a unique value of deviatoric strain energy, $\varepsilon_{v,liq}$, for specific values of e_0 (or Dr) Sr_0 , for different granular soils with $Sr \leq 100\%$. Hence, the relationship between p'/p'_0 , and $\varepsilon_v/\varepsilon_{v,liq}$ in undrained cyclic tests does not depend on intrinsic properties and state of the soil, the behaviour follows the general law of Eq. 23 (also represented in

Table 3
Studies on the liquefaction resistance of partially saturated sands.

Reference	Interpretation approach	Sr tested	Evaluation of liquefaction resistance
(Yoshimi et al. (1989)	Stress-based	70 % to 100 %	B-value decreases; N_L increases
Tsukamoto et al. (2002)	Stress- and wave based	90 % to 100 %	B-value decreases; V_p increases; N_L increases
Okamura and Soga (2006)	Stress-based	70 % to 100 %	S_r decreases; ε_v increases; N_L increases
Yegian et al. (2007)	Stress-based	72 % to 100 %	S_r decreases; r_u decreases
Unno et al. (2008)	Stress-based	12 % to 100 %	s increases; N_L increases
Eseller-Bayat et al. (2013)	Stress-based	40 % to 100 %	S_r decreases; γ decreases; r_u decreases
He et al. (2014)	Stress-based	84 % to 100 %	S_r decreases; γ decreases; r_u decreases
Vega-Posada et al. (2014)	Stress-based	83 % to 100 %	S_r decreases; USR increases; r_u decreases
Tsukamoto et al. (2014)	Stress-based	42 % to 100 %	s increases; ε_v increases; N_L increases
Marasini and Okamura (2015)	Stress-based	85 % to 100 %	S_r decreases; r_u decreases
Wang et al. (2016)	Stress-based	68 % to 100 %	S_r decreases; ε_v increases; N_L increases
O'Donnell et al. (2017)	Stress-based	82 % to 100 %	S_r decreases; V_p decreases; r_u decreases; N_L increases
Zeybek and Madabhushi (2017)	Stress-based	85 % to 100 %	S_r decreases; r_u decreases
Mele et al. (2019)	Energy-based	54 % to 100 %	S_r decreases; ε_v increases; W_{tot} increases; N_L increases
Tsukamoto (2019)	Stress-based	41 % to 100 %	S_r decreases; USR increases; N_L increases
Ishihara et al. (2020)	Stress-based	90 % to 100 %	B-value decreases; V_p increases; N_L increases
Amanta and Dasaka (2021)	Stress-based	41 % to 100 %	S_r decreases; N_L increases
Mousavi and Ghayoomi (2021)	Stress-based	79 % to 100 %	S_r decreases; r_u decreases
Wang et al. (2021)	Stress-based	83 % to 100 %	S_r decreases; USR increases; N_L increases
Banerjee et al. (2022)	Stress-based	63 % to 100 %	s increases; N_L increases
Chavan et al. (2021)	Stress-based	70 % to 100 %	B-value decreases; N_L increases
Mele et al. (2022)	Energy-based	72 % to 100 %	S_r decreases; ε_v increases; W_{tot} increases; N_L increases
Zeybek (2022a)	Stress-based	85 % to 100 %	S_r decreases; r_u decreases
Fardad Amini and Yang (2023)	Stress- and wave based	90 % to 100 %	B-value decreases; V_p/V_s decreases; N_L increases
Molina-Gómez et al. (2023a)	Stress- and wave based	91 % to 100 %	S_r decreases; V_p decreases; V_s constant; r_u decreases; N_L increases
Molina-Gómez et al. (2023b)	Stress- and wave based	82 % to 100 %	S_r decreases; V_p decreases; V_p/V_s decreases; r_u decreases; N_L increases
Zuo et al. (2024)	Stress- and wave based	80 % to 100 %	B-value decreases; V_p increases; N_L increases
Fardad Amini and Yang (2024)	Energy-based	90 % to 100 %	B-value decreases; ΔW increases; N_L increases

Note: S_r is degree of saturation; N_L is the number of cycles required to liquefaction; V_p is P-wave velocity; V_s is S-wave velocity; ε_v is volumetric strain; r_u is pore pressure ratio; s is the matric suction; γ is shear strain; W_{tot} total energy; ΔW dissipated energy; and USR is undrained shear resistance.

Fig. 13).

$$\frac{p'}{p_0} = 1 - \left(\frac{\varepsilon_v}{\varepsilon_{v,liq}} \right)^{1.7} \quad (23)$$

4. Analysis of experimental data and discussion

Experimental studies on the liquefaction resistance of granular soils with varying S_r have been extensively reported in the literature, encompassing element tests and physical models subjected to static and dynamic loads. This section compiles experimental results on the liquefaction resistance of various sands tested across a wide range of S_r values and analyses these results using stress-based approaches. In addition, this analysis is complemented by introducing a novel interpretation involving wave-based approaches, which can be utilised for further comparisons between field and laboratory testing.

Table 3 provides an overview of experimental studies assessing the liquefaction resistance of desaturated sands using several interpreting approaches. These studies were conducted in the laboratory under controlled conditions and in cyclic/monotonic triaxial, torsional shear or centrifuge apparatuses. In these tests, some authors monitored the evolution of S_r in the soil during the backpressure ramp increment by measuring Skempton's pore pressure parameter B (B -value) or P-wave velocity (V_p) (Kumar and Madhusudhan, 2012; Leong and Cheng, 2016; Astuto et al., 2022). The V_p was estimated by compressional disks (CD) and bender element (BE) tests. In some of the studies in Table 3, different desaturation techniques were applied, e.g. air/gas injection, denitrification, water electrolysis or control of the back pressure during testing. The V_p results allow the laboratory results to be linked to field measurements (Amoroso et al., 2020). Other authors monitored the water volume in the soil sample to estimate the S_r based on the phase relationships and the soil water characteristic curve (Mirshekari and Ghayoomi, 2017; Mousavi et al., 2021). The findings of Table 3 demonstrate the potential of reducing S_r for mitigating the earthquake-induced liquefaction and the plausibility of applying different interpretation methods to assess the undrained cyclic behaviour of desaturated sands with diverse S_r .

Fig. 14 contains the results of experimental studies comparing the cyclic liquefaction resistance of well-characterised sands at different degrees of saturation, including the full-saturation condition. The results in Fig. 14 confirm the evolution of the cyclic behaviour at different S_r and indicate that the liquefaction resistance increases with decreasing S_r for different sands.

Fig. 15 illustrates the cyclic behaviour observed in three cyclic triaxial (CTx) tests conducted under the same state parameter but with different S_r in TP-Lisbon sand (Molina-Gómez et al., 2023b). The results of the CTx tests under similar cyclic loading conditions (with CSR between 0.152 and 0.158) highlight significant differences in the cyclic behaviour of the specimens when S_r varies for 100 %, 96 %, and 91 %. In particular, the test performed with $S_r = 91$ % exhibited lower axial strains (ε_a) compared to the other tests, while specimens tested under fully saturated conditions demonstrated the highest deformation after onset of liquefaction. This behaviour suggests that partial saturation attenuates the liquefaction-induced deformations, which is consistent with findings reported by Zeybek and Madabhushi (2016). However, despite the degree of saturation, all CTx tests showed significant deformations (i.e. $\varepsilon_a > 5$ %) after the onset of liquefaction. Furthermore, Fig. 15 illustrates the evolution of the pore pressure coefficient ($r_u = \Delta u/p'_0$). This representation shows values higher than 0.99, even for fully and partially saturated specimens. Consequently, the results of Fig. 15 clearly demonstrate that liquefaction resistance increases as the soil becomes desaturated, which is due to the reduction in pore pressure build-up caused by the presence of occluded air bubbles in the pore fluid (Yegian et al., 2007).

Comparison of the results in Fig. 15 for similar test conditions indicates that a 4 % reduction in S_r triples the number of cycles to the

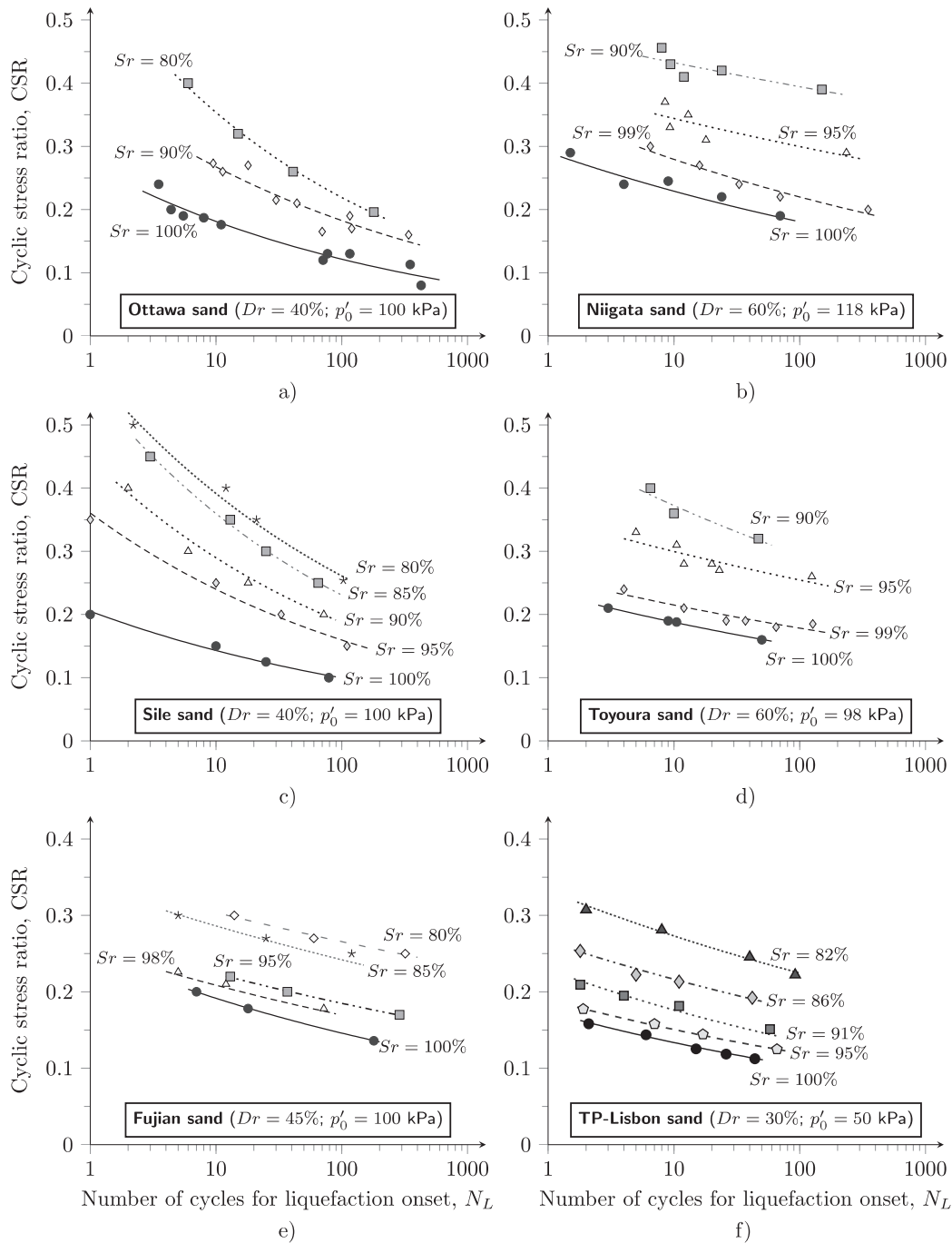


Fig. 14. Influence of degree of saturation on the liquefaction resistance: a) Ottawa sand (after Sherif et al., 1977); b) Niigata sand (after Ishihara et al., 2020); c) Silé sand (after Zeybek, 2022a); d) Toyoura sand (after Tsukamoto et al., 2002); e) Fijian sand (after Zuo et al., 2024); f) TP-Lisbon sand (after Molina-Gómez et al., 2023b).

onset of liquefaction. Similarly, a 9 % reduction in S_r significantly increased the liquefaction resistance from 2 cycles to 58 cycles. However, the representation in Fig. 15 does not quantify the magnitude of the improvement in the liquefaction resistance.

Therefore, the results were further interpreted using the liquefaction resistance ratio (LRR) to explain such an improvement. The LRR normalises the liquefaction resistance for different S_r with the liquefaction resistance reported under fully saturated conditions (Tsukamoto et al., 2002; Ishihara et al., 2020). The LRR was calculated by computing the ratio between the CSR corresponding to a specific number of cycles for liquefaction onset, N_L under partially saturated conditions (i.e. CSR_g for $S_r < 100\%$) and the CSR corresponding to the number of cycles of

liquefaction onset under fully saturated conditions (i.e. CSR_g for $S_r = 100\%$). Typically, N_L equal to 15 and 20 describe earthquake-induced liquefaction (Ishihara, 1993), leading to LRR_{15} and LRR_{20} , respectively.

Fig. 16 shows the LRR as a function of S_r for the data of the five sands analysed in Fig. 14. A series of linear relationships can be derived from experimental data to describe the evolution of liquefaction resistance under different S_r in such sands. The results in Fig. 16 indicated that reducing S_r from 100 % to 90 % increases the liquefaction resistance by at least 1.5 times, providing experimental evidence for the effectiveness of soil desaturation or induced partial saturation as a technique to mitigate earthquake-induced liquefaction.

Although there is strong evidence for the excellent performance of

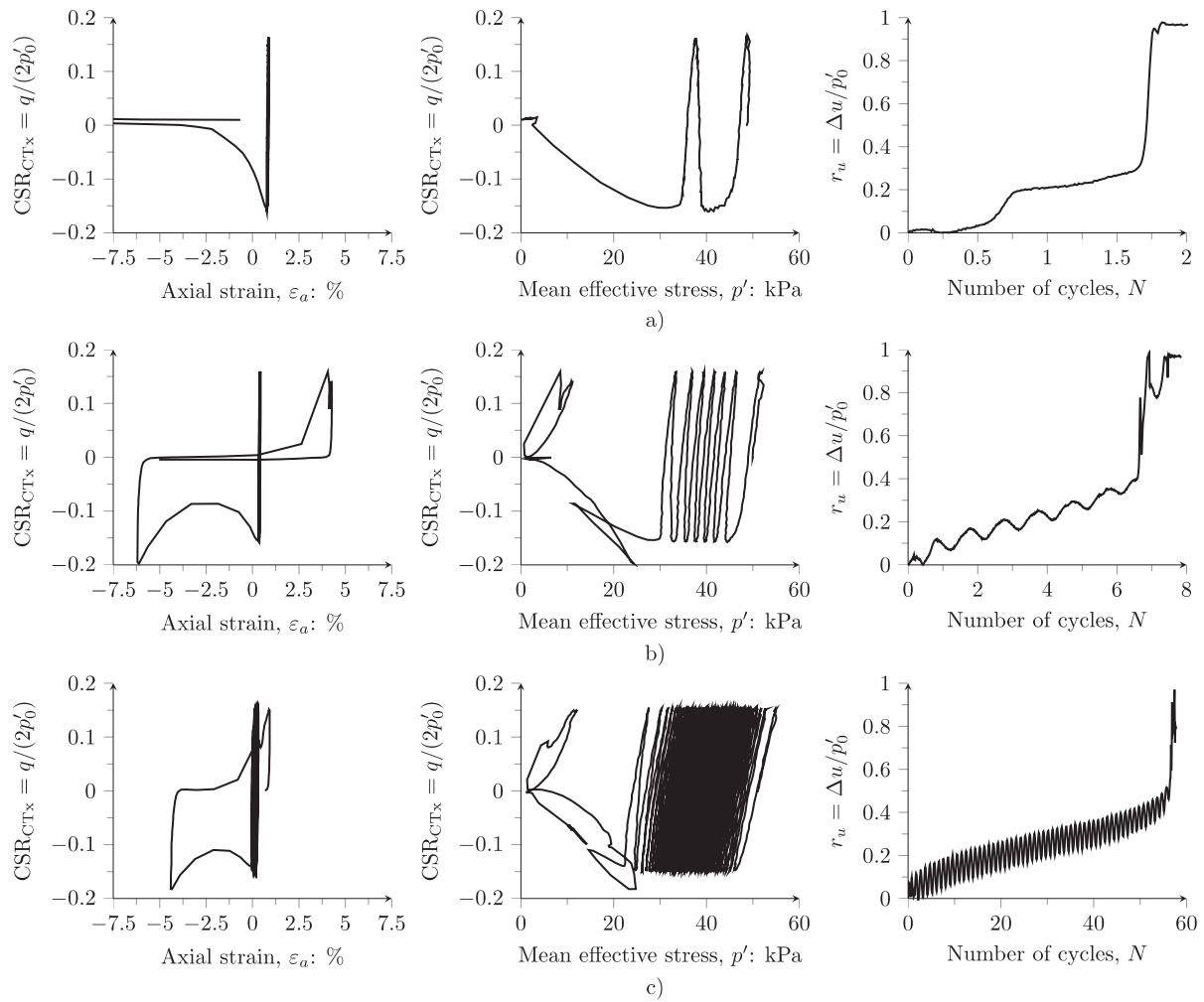


Fig. 15. Undrained cyclic behaviour under different S_r (after Molina-Gómez et al., 2023b): a) $S_r = 100\%$; b) $S_r = 95\%$; c) $S_r = 91\%$.

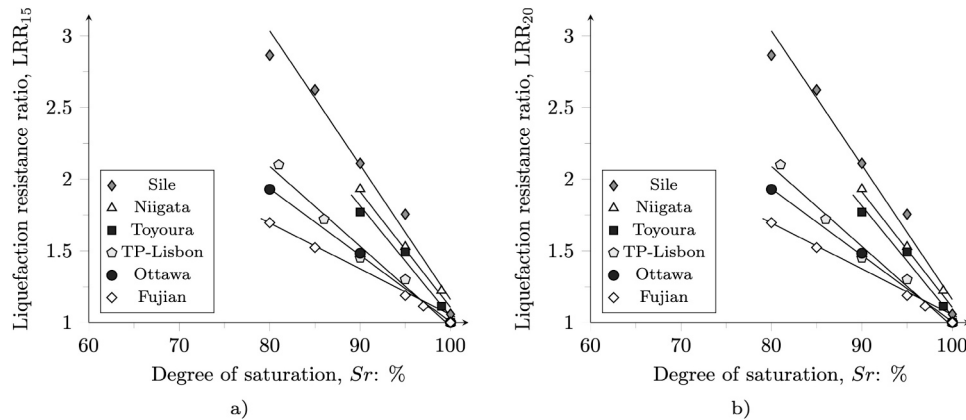


Fig. 16. Liquefaction resistance ratio for different sands and S_r – data from Sherif et al. (1977); Ishihara et al. (2020); Tsukamoto et al. (2002); Zeybek (2022a); Molina-Gómez et al. (2023b): a) LRR for 15 cycles; b) LRR for 20 cycles.

soil desaturation, the linear trends describing the improvement in liquefaction resistance do not converge to a common factor. These differences may be attributed to the test conditions in each study or to the soil state (i.e. Dr and p'_0), which affects the cyclic behaviour of the sands due to their influence on soil stiffness. A very similar behaviour was observed for LRR estimated for 15 and 20 cycles for the onset of liquefaction. This behaviour can be attributed to the parallel trends reported

in Fig. 14 for each S_r for all sands.

Because of this, Fig. 17 presents an alternative approach to describe the improvement of liquefaction resistance, which could capture the influence of S_r on the cyclic behaviour of sands. This approach relates the improvement of liquefaction resistance by changing S_r to V_p measurements on natural sands that have experienced liquefaction in Japan and Portugal and characterised by combined CTx and BE tests (Ishihara

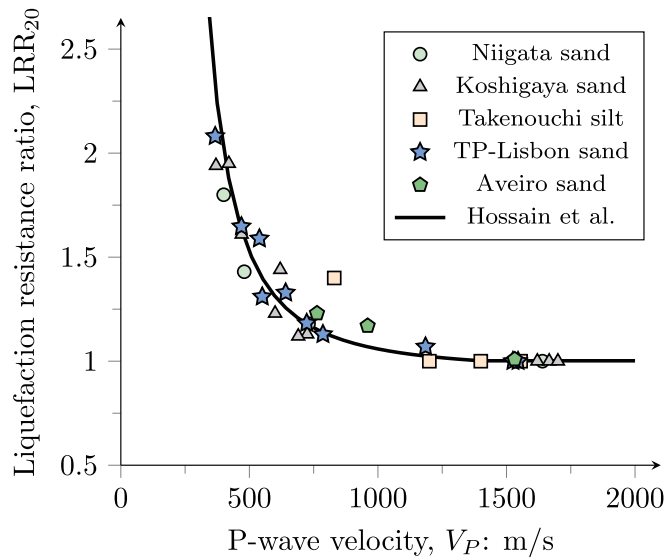
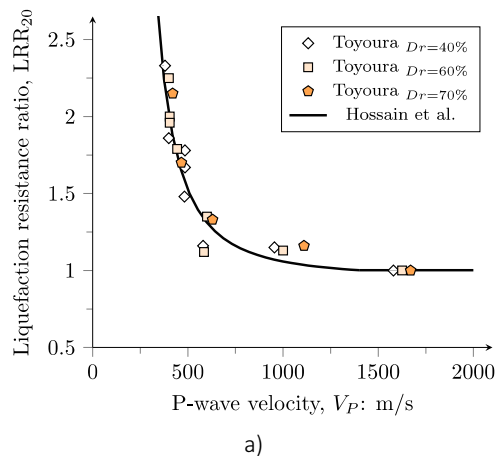


Fig. 17. LRR_{20} and V_P relationship for liquefiable Japanese and Portuguese sands – data from Ishihara et al. (2020); (Cordeiro et al., 2022); Molina-Gómez et al. (2023b).

et al., 2020; Cordeiro et al., 2022; Molina-Gómez et al., 2023b). V_P is a reliable estimator of S_r in sands, as shown in Fig. 4. The liquefaction resistance of soil can be assessed using the CSR, which triggers liquefaction after a specified number of load cycles, e.g. 15 or 20 cycles. In this approach, the CSR at the 20th loading cycle on the CSR vs. N_L curve is used to determine the liquefaction resistance. The reduction in S_r is quantified by the liquefaction resistance ratio at 20 cycles (LRR_{20}), which is the ratio between the CSR at 20 cycles for any S_r lower than 100 % and the CSR at 20 cycles for fully saturated conditions (i.e. $S_r = 100$ %). The experimental data in Fig. 17 follow a non-linear trend, in which all experimental data are equal to 1 for V_P values above 1500 m/s (corresponding to $S_r = 100$ %). Furthermore, the results for LRR_{20} indicate a significant increase in V_P of approximately 500 m/s, which corresponds to a notable rise in liquefaction resistance for sands with V_P values below 500 m/s. These results fit with the equation proposed by Hossain et al. (2013):

$$LRR_{20} = 0.95 + \exp \left[2 - 2.95 \ln \left(\frac{V_P}{100} + 1.3 \right) \right] \quad (24)$$

Fig. 17 provides a reliable estimator for the improvement of liquefaction resistance due to the dependence of P-wave propagation on S_r (Santamarina et al., 2005; Ishihara et al., 2020; Astuto et al., 2022).



These results demonstrate the effectiveness of using P-wave measurements to indirectly monitor the increase in the cyclic resistance of desaturated sands. This improvement is due to the reduction in the bulk modulus of the pore fluid – caused by the presence of gas/air bubbles – which reduces the pore pressure build-up generated by reducing the soil volume during cyclic loading (Tsukamoto et al., 2002; Molina-Gómez et al., 2023b). However, to confirm the non-linear fit shown in Fig. 17, further experimental results from different granular soils were required, specially from sandy soils with different relative densities (Dr) or different fines content (FC), so Fig. 18 investigates the effects of Dr and FC on the wave-based approach to evaluate the improvement of liquefaction resistance.

Fig. 18 reveals that a single relationship between LRR_{20} and V_P can capture the effects of Dr and FC. This confirms that the equation proposed by Hossain et al. (2013) describes well the liquefaction resistance improvement in different sands under several test conditions using a P-wave-based approach. In addition, the results of the wave-based approach revealed that soils with $S_r \approx 90$ % or $V_P \approx 460$ m/s have a cyclic resistance about twice as high as those sands with $S_r = 100$ %. The improvement in liquefaction resistance in desaturated sands is caused by occluded air bubbles, and it can be detected by the reduction in the P-wave propagation. This confirms the S_r effects on the liquefaction resistance, where it is plausible that a reduction in S_r leads to an increase in the cyclic resistance of liquefiable soils, demonstrating its great potential for earthquake-induced liquefaction mitigation.

5. Summary and concluding remarks

The desaturation or induced partial saturation has garnered increasing interest as an alternative technique for liquefaction mitigation due to the gas generation in fully saturated soils. Gas bubbles increase fluid compressibility and significantly reduce pore pressure build-up during cyclic loading. This paper has explored the efficiency of desaturation for earthquake-induced liquefaction and introduced novel approaches to assess the cyclic behaviour of sands under diverse degrees of saturation. The analysis performed herein led to the following conclusions:

- Experimental evidence shows that even a slight decrease in the degree of saturation can significantly improve the earthquake-induced liquefaction resistance of sands. This effect emphasises the potential of reducing the degree of saturation to mitigate earthquake-induced liquefaction since the experimental evidence presented herein shows that the reduction in the degree of saturation increases the number of cycles for liquefaction onset and mitigates the strains generated during cyclic loading.

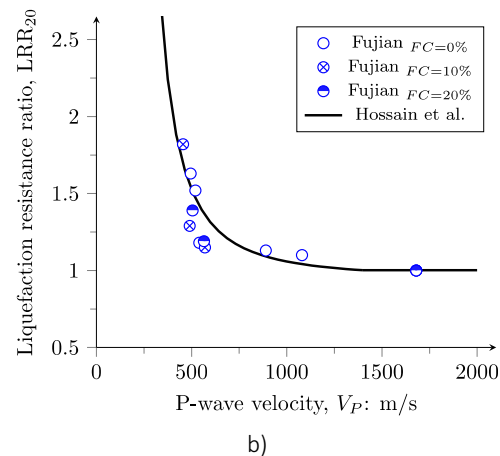


Fig. 18. LRR_{20} and V_P relationship: a) Effects of Dr –data from Tsukamoto et al. (2002); b) Effects of FC –data from Zuo et al. (2024).

- Energy-based methods emerged as promising tools for evaluating the cyclic behaviour of granular soils across varying degrees of saturation. Unlike conventional stress-based methods, energy-based approaches account for fluid compressibility effects on soil behaviour. Moreover, these methods facilitate the comparison of results obtained from diverse laboratory testing procedures, including element testing and centrifuge modelling.
- Experimental data reported in various sands showed that the liquefaction strength curves are parallel and move up as S_r decreases. However, the improved liquefaction resistance provided by the S_r reduction does not follow a unique trend, as observed by estimating the relationship between the normalised cyclic strength and S_r in different sands. This divergence is attributed to variations in confinement pressure, relative density and initial fabric of the soil. Further research varying these parameters is necessary to characterise the factors affecting the undrained cyclic behaviour in sands with diverse S_r .
- A wave-based approach proved the potential of P-wave velocity to describe the liquefaction resistance improvement by reducing S_r . This approach addresses an equation that correlates the LRR_{20} with V_p for various liquefiable sands. Experimental data shows that this approach aligns well for sands with different D_r and FC, indicating that the LRR_{20} : V_p relationship is unaffected by the testing conditions. However, wave-based approaches are more limited due to the scarcity of comprehensive laboratory data that includes cyclic element testing with wave measurements under different S_r conditions, suggesting a novel possibility for future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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